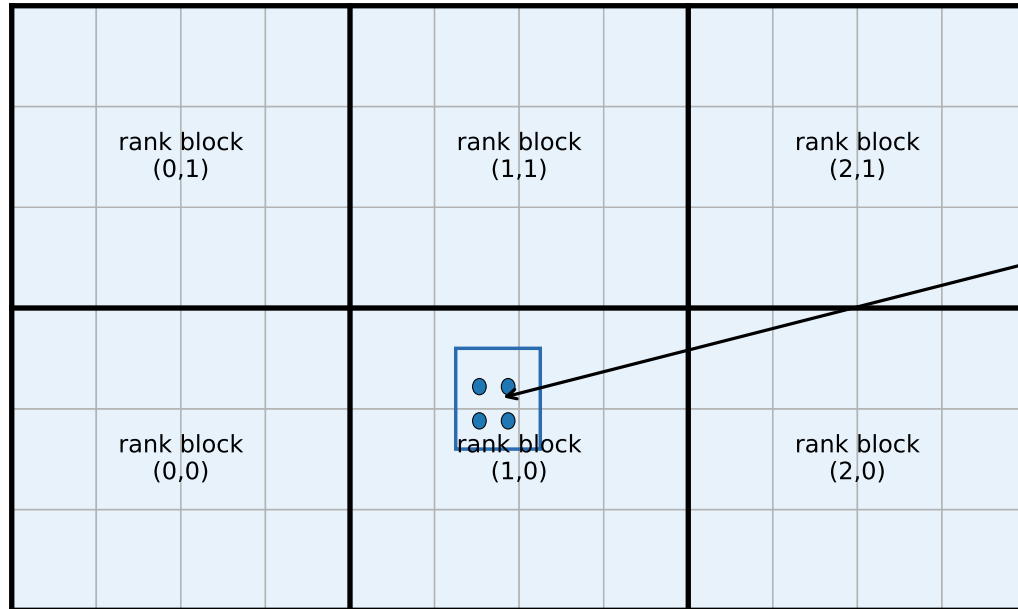


uniform InternalMesh cells; equal cells per partition

$xpar = refine * base_xpar$
(periodic directions usually ≥ 3)

ghost padding
(non-PBC)



cpp cells per partition

Plane strain

$zpar = 1, nK = 3$

one particle layer through thickness

periodic z is disabled internally

$ppc = 2$ in each
active direction
(4 particles/cell in 2D)

Common PFW resolution pattern

cpp = cells per partition

$refine$ = resolution scale

$xpar$ = $refine * base_xpar$

nI = $xpar * cpp$ (plus ghost/PBC logic)

$ppcx$ = ppc (or direction-specific)

Uniform refinement in d dimensions increases particles/ranks as r^d and reduces stable explicit Δt approximately as $1/r$.